

U.G.C. NET Exam

COMPUTER

(Solved with Expl)

1. A file is downloaded in a home computer using a 56 kbps MODEM connected to an Internet Service Provider. If the download of file completes in 2 minutes, what is the maximum size of data downloaded?
- (a) 112 Mbits (b) 6.72 Mbits
(c) 67.20 Mbits (d) 672 Mbits

Ans : (b) We can download 56 kbits in one second. Then data download in 2 minutes.

$$\begin{aligned} &= 56 \text{ K} \times 2 \times 60 \\ &= 56 \times 120 = 6720 \text{ K bits} \\ &= 6.72 \text{ M bits} \end{aligned}$$

2. In _____ CSMA protocol, after the station finds the line idle, it sends or refrains from sending based on the outcome of a random number generator.
- (a) Non-persistent (b) 0-persistent
(c) 1-persistent (d) p-persistent

Ans : (d) In P -persistent CSMA protocol, when station found channel idle, then it send or refrain from sending based on probability P .

3. Which of the following substitution technique have the relationship between a character in the plaintext and a character in the ciphertext as one-to-many?
- (a) Monoalphabetic (b) Polyalphabetic
(c) Transpositional (d) None of these

Ans : (b) Substitution ciphers encrypt plaintext by replacing one symbol with another. In polyalphabetic substitution, each occurrence of a character may have a different substitute. Thus, the relationship between a character in the plaintext to a character in the ciphertext is one too many. While, in monoalphabetic, substitution, the relationship between a symbol in the plaintext to a symbol in the ciphertext is always one-to-one.

4. What is the maximum length of CAT-5 UTP cable in Fast Ethernet network?
- (a) 100 meters (b) 200 meters
(c) 1000 meters (d) 2000 meters

Ans : (a) Category 5 cable, commonly referred to as cat 5, is a twisted pair cable for carrying signals. This type of cable is used in structured cabling for computer network such as ethernet. The cable standard provides performance of up to 100 MHz and is suitable for 10Base-T, 100Base-TX (fast ethernet), 1000Base-T (Gigabit ethernet) and 2.5GBase-T. cat 5 is also used telephony and video. The maximum length for a cable segment is 100 m per TIA/EIA 568-5-A.

n. September, 2013

R SCIENCE

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5. The _____ is a set of standards that defines how a dynamic web document should be written, how input data should be supplied to the program, and how the output result should be used.
- (a) Hyper Text Markup Language
 - (b) File Transfer Protocol
 - (c) Hyper Text Transfer Protocol
 - (d) Common Gateway Interface

Ans : (d) 1. Common Gateway Interface (CGI) is a technology that creates and handles dynamic documents. CGI is a set of standards that defines how a dynamic document should be written, what input data should be supplied to the program, and how the output result should be used.

2. HTML (Hypertext markup Language) is a markup language.

3. FTP (file transfer protocol) is a standard application layer protocol used to transfer computer files between a client and server on a computer network.

4. The HTTP (hypertext transfer protocol) is an application layer protocol for distributed, collaborative, hypermedia information systems.

6. The count-to-infinity problem is associated with
- (a) Flooding algorithm
 - (b) Hierarchical routing algorithm
 - (c) Distance vector routing algorithm
 - (d) Link state routing algorithm

Ans : (c) The Bellman-Ford algorithm does not prevent routing loops from happening and suffers from the count-to-infinity problem.

Distance-vector routing protocols use the Bellman-ford algorithm and link-state routing protocols uses the dijkstra's algorithm.

7. The IEEE single-precision and double-precision format to represent floating-point numbers, has a length of _____ and _____ respectively.
- (a) 8 bits and 16 bits
 - (b) 16 bits and 32 bits
 - (c) 32 bits and 64 bits
 - (d) 64 bits and 128 bits

Ans : (c) 1. Single precision floating point format is a computer format that occupies 4 bytes (32 bits) in computer memory and represents a wide, dynamic range of values by using a floating point.

2. Double precision floating point format is a computer number format that occupies 8 bytes (64 bits) in computer memory and represents a wide, dynamic range of values by using a floating point.

8. Consider an undirected graph G with 100 nodes. The maximum number of edges to be included in G so that the graph is not connected is

- (a) 2451 (b) 4950
(c) 4851 (d) 9900

Ans : (c) Maximum number of edges in a simple graph

$$e = \frac{(n-k)(n-k+1)}{2}$$

where, k is the number of connected component (i.e., if graph is connected graph then $k=1$).

Since, given graph is disconnected, so, we take $k=2$ for maximum number of edges.

$$e = \frac{(n-1)(n-2)}{2}$$

we assume $k_{(n-1)}$ complete graph and other one vertex.

Therefore,

$$e = \frac{(n-1)(n-2)}{2} = \frac{99 \times 98}{2} = 4851$$

9. The amortized time complexity to perform _____ operation (s) in Splay trees is $O(\lg n)$.

- (a) Search
(b) Search and Insert
(c) Search and Delete
(d) Search, Insert and Delete

Ans : (d) A **splay tree** is a self-adjusting binary search tree with the additional property that recently accessed elements are quick to access again. It performs basic operations such as insertion, look-up and removal in $O(\lg n)$ amortized time.

10. Suppose that the splits at every level of Quicksort are in proportion $1-\beta$ to β , where $0 < \beta < 0.5$ is a constant. The number of elements in an array is n. The maximum depth is approximately

- (a) $0.5 \beta \lg n$ (b) $0.5 (1-\beta) \lg n$
(c) $-(\lg n)/(\lg \beta)$ (d) $-(\lg n)/\lg(1-\beta)$

Ans : (d) First calculated in the maximum height.

$$\Rightarrow n(1-\beta)^x \leq 1$$

$$\Rightarrow x \geq \log_{(1-\beta)} \left(\frac{1}{n} \right)$$

$$\Rightarrow x \geq -\frac{\log(n)}{\log(1-\beta)}$$

Note that we can also find minimum height replace $(1-\beta)$ to β .

- 11. The minimum number of nodes in a binary tree of depth d (root is at level 0) is**
- (a) $2^d - 1$ (b) $2^{d+1} - 1$
 (c) $d + 1$ (d) d

Ans : (b) In a binary tree, you have 1 root, 2 sons of that root, 4 grandsons, 8 grand-grandsons, and so on. So, the total number of nodes is the sum of the geometric series.

$$1 + 2 + 4 + 8 + \dots + 2^d = \frac{2^{d+1} - 1}{2 - 1} = 2^{d+1} - 1$$

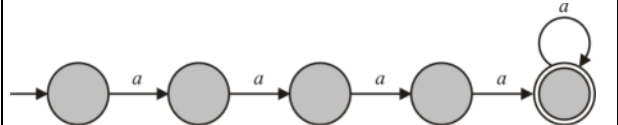
where, d is the depth (i.e., for $d = 0$, we have 1 node).

- 12. The efficient data structure to insert/delete a number is a stored set of numbers is**
- (a) Queue (b) Linked list
 (c) Doubly linked list (d) Binary tree

Ans : (c) Among given data structure double linked list is best data structure since we can traverse in any direction. Searching take $O(n)$ and insertion and deletion take $O(1)$ time.

- 13. The number of states in a minimal deterministic finite automaton corresponding to the language $L = \{a^n | n \geq 4\}$ is**
- (a) 3 (b) 4
 (c) 5 (d) 6

Ans : (c) Given, language is $L = \{a^n | n \geq 4\}$ minimal deterministic finite automaton is



Total number of states is 5.

- 14. Regular expression for the language $L = \{W \in \{0, 1\}^* | W \text{ has no pair of consecutive zeros}\}$ is**
- (a) $(1 + 010)^*$ (b) $(01 + 10)^*$
 (c) $(1 + 010)^*(0 + \lambda)$ (d) $(1 + 01)^*(0 + \lambda)$

Ans : (d)

(a) $(1 + 010)^* = \{\epsilon, 1, 010, 1010, 0101, 010010, 11, \dots\}$
 string 010010 contains pair of consecutive zeros.

(b) $(01 + 10)^* = \{\epsilon, 01, 10, 0110, 1001, \dots\}$
 string 1001 contains pair of consecutive zeros.

(c) $(1 + 010)^*(0 + \epsilon) = \{\epsilon, 1, 010, 010010, \dots\}$
 string 010010 contains pair of consecutive zeros.

(d) $(1 + 01)^*(0 + \epsilon) = \{\epsilon, 1, 0, 01, 10, 11, \dots\}$
 it can not have any string with pair of consecutive zeros.

- 15. Consider the following two languages:**
- $L_1 = \{a^n b^l a^k | n + l + k > 5\}$
 $L_2 = \{a^n b^l a^k | n > 5, l > 3, k \geq l\}$

Which of the following is true?

- (a) L_1 is regular language and L_2 is not regular language.
- (b) Both L_1 and L_2 are regular languages.
- (c) Both L_1 and L_2 are not regular languages.
- (d) L_1 is not regular language and L_2 is not regular language.

Ans : (a) Given, $L_1 = \{a^n b^l a^k \mid n+l+k > 5\}$ we can write L_1 as given below

$L_1 = \{a^* b^* c^*\} \cap \{|w| > 5 \mid w \in \{a, b\}^*\}$ we can draw DFA for L_1 , so it is regular.

Given $L_2 = \{a^n b^l c^k \mid n > 5, l > 3, k \leq l\}$ language L_2 can not be regular since it has infinite comparison (i.e., $k \leq l$) but, L_2 is context free language, since, we can identify any string of L_2 using only one stack.

16. LL grammar for the language $L = \{a^n b^m c^{n+m} \mid m \geq 0, n \geq 0\}$ is

- (a) $S \rightarrow aSc \mid S_1; S_1 \rightarrow bS_1c \mid \epsilon$
- (b) $S \rightarrow aSc \mid S_1 \mid \epsilon; S_1 \rightarrow bS_1c$
- (c) $S \rightarrow aSc \mid S_1 \mid \epsilon; S_1 \rightarrow bS_1c \mid \epsilon$
- (d) $S \rightarrow aSc \mid \lambda; S_1 \rightarrow bS_1c \mid \epsilon$

Ans : (c) Given language is:

$$L = \{a^n b^m c^{n+m} \mid m, n \geq 0\}$$

$$L = \{a^n b^m c^m c^n \mid m, n \geq 0\}$$

Context free grammar for L is

$$S \rightarrow aSc \mid \epsilon \mid S_1$$

$$S_1 \rightarrow bS_1c \mid \epsilon$$

17. Assume the statements S_1 and S_2 given as:

S_1 : Given a context free grammar G , there exists an algorithm for determining whether $L(G)$ is infinite.

S_2 : There exists an algorithm to determine whether two context free grammars generate the same language.

Which of the following is true?

- (a) S_1 is correct and S_2 is not correct.
- (b) Both S_1 and S_2 are correct.
- (c) Both S_1 and S_2 are not correct.
- (d) S_1 is not correct and S_2 is correct.

Ans : (a) **S_1** : Given a context-free grammar $G = (V, T, S, P)$ there exists an algorithm for determining whether or not $L(G)$ is infinite.

Since, we have an algorithm for deciding whether a grammar has a repeating variable, we have an algorithm for determining whether or not $L(G)$ is infinite.

S_2 : There does not exist any algorithm to determine whether two context free grammars generate the same language since, it is impossible to even decide whether a CFG is equivalent to the trivial CFG defining the language of all strings.

18. The number of eight-bit strings beginning with either 111 or 10 is ____.
- (a) 64 (b) 128
(c) 265 (d) None of these

Ans : (a) Total number of 8 bit string starting with 111.

1	1	1					
---	---	---	--	--	--	--	--

Remaining five places we can fill either 0 or 1, (i.e., 2 choices are available). Then number of strings = $2^5 = 32$.

Similarly, total number of 8 bit strings starting with 101 = $2^5 = 32$.

So, total number of strings = $32 + 32 = 64$.

19. Find the number of ways to paint 12 office so that 3 of them will be green, 2 of them pink, 2 of them yellow and the rest ones white.
- (a) 55,440 (b) 1,66,320
(c) 4790E+08 (d) 39,91,680

Ans : (b) Total number of ways 12 offices can be painted = $12!$. But 3 of them will be green 2 of them pink, 2 of them yellow and remaining 5 to them will be white.

Therefore, using permutations of multi-sets

$$= \frac{12!}{3! \times 2! \times 2! \times 5!} = 166320$$

20. Consider the following statements:
- (i) A graph in which there is a unique path between every pair of vertices is a tree.
(ii) A connected graph with $e = v - 1$ is a tree.
(iii) A graph with $e = v - 1$ has no circuit is a tree.
- Which of the above statement is/are true?
- (a) (i) and (iii) (b) (ii) and (iii)
(c) (i) and (ii) (d) All of these

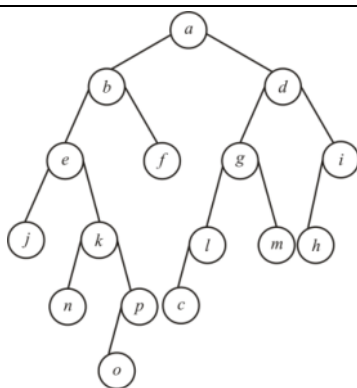
Ans : (d) A tree is an undirected graph in which any two vertices are connected by exactly one path (i.e., minimally connected graph) with $e = (n - 1)$ edges. Therefore, all statements are true.

21. Consider the in-order and Post-order traversals of a tree a given below:
- In-order:** *J e n k o p b f a c l g m d h i*
Post-order: *j n o p k e f b c l m g h i d a*
The pre-order traversal of the tree shall be
- (a) *a b f e j k n o p c d g l m h i*
(b) *a b c d e f j k n o p g l m h i*
(c) *a b e j k n o p f c d g l m h i*
(d) *j e n o p k f b c l m g h i d a*

Ans : (*) We assume that given in order and postorder traversal of a binary tree.

Inorder: *j e n k o p b f a c l g m d h i*
Postorder: *j n o p k e f b c l m g h i d a*

Construction of binary tree using in order and postorder:



Therefore, preorder is:

a b e j k n p o f d g l c m i h

But none option is matched.

22. A simple graph G with n-vertices is connected if the graph has

- (a) $(n-1)(n-2)/2$ edges
- (b) more than $(n-1)(n-2)/2$ edges
- (c) less than $(n-1)(n-2)/2$ edges
- (d) $\sum_{i=1}^k C(n_i, 2)$ edges

Ans : (b) Let e be the number of edges n the number of vertices and k the number of connected component of a graph G . Then maximum number of edges are

$$e \leq \frac{(n-k)(n-k+1)}{2}$$

For connected graph maximum number of edges will be as edges of complete graph k_n (when connected component $k = 1$), then

$$e \leq \frac{n(n-1)}{2}$$

For disconnected graph maximum number of edges will be as edges of complete graph $k_{(n-1)}$ and single vertex (when connected component $k = 2$).

$$e \leq \frac{(n-1)*(n-2)}{2}$$

Therefore, graph to be connected graph has more than $(n-1)*(n-2)/2$ edges.

23. Which one of the following set of gates is best suited for 'parity' checking and 'parity' generation?

- (a) AND, OR, NOT
- (b) NAND, NOR
- (c) EX-OR, EX-NOR
- (d) None of these

Ans : (c) A **parity bit or check bit** is a bit added to a string of binary code that indicates whether the number of 1 bits in the string is even or odd. Parity bits are used as the simplest form of error detecting code. There are two variants of parity bits: even parity bit and odd parity bit. Parity has the advantage that it uses only a single bit and requires only a number of XOR gates to generate. The parity generate and parity checker can

both be built using XOR gates. To generate odd parity, simply invert the even parity. The Ex-OR (XOR) gate can be EXNOR gate.

24. The quantification $\exists!xP(x)$ denotes the proposition "There exists a unique x such the $P(x)$ is true". express the quantification using universal and existential quantification and logical operations:

- (a) $\exists xP(x) \vee \forall x\forall y((P(x) \vee P(y) \rightarrow x = y)$
- (b) $\forall xP(x) \wedge \forall x\forall y((P(x) \vee P(y) \rightarrow x = y)$
- (c) $\exists xP(x) \wedge \forall x\forall y((P(x) \wedge P(y) \rightarrow x = y)$
- (d) $\exists xP(x) \wedge \exists x\exists y((P(x) \vee P(y) \rightarrow x = y)$

Ans:(c)

$$\exists!x(P(x)) = [\exists x P(x)] \wedge \forall x \forall y [P(x) \wedge P(y) \rightarrow x = y]$$

On LHS, hence the $\exists!xP(x)$ first clauses says there is at least one, $\forall x \forall y [P(x) \wedge P(y) \rightarrow x = y]$ second says there is at most one x , x for which $P(x)$ is true.

25. If F and G are Boolean functions of degree n . Then, which of the following is true?

- (a) $F \leq F + G$ and $FG \leq F$
- (b) $G \leq F + G$ and $FG \geq G$
- (c) $F \geq F + G$ and $FG \leq F$
- (d) $G \geq F + G$ and $FG \leq F$

Ans : (b) Given F and G are boolean functions of degree n , i.e., number of boolean functions.

$$F = 2^{2^n} = G$$

Therefore,

$$\Rightarrow G \leq F + G$$

$$\Rightarrow 2^{2^n} \leq 2^{2^n} + 2^{2^n}$$

$$\Rightarrow 2^{2^n} \leq 2(2^{2^n}) \text{ is true.}$$

and also,

$$\Rightarrow G \leq FG$$

$$\Rightarrow 2^{2^n} \leq (2^{2^n}).(2^{2^n})$$

$$\Rightarrow 2^{2^n} \leq 2^{(2^n + 2^n)} \text{ is also true.}$$

26. Match the following identities/laws to their corresponding name:

- | List-1 | List-II |
|-------------------------------------|---------------|
| A. $x + x = x$
$x \bullet x = x$ | 1. Dominance |
| B. $x + 0 = x$
$x \bullet 1 = x$ | 2. Absorption |
| C. $x + 1 = 1$
$x \bullet 0 = 0$ | 3. Idempotent |
| D. $x \bullet (x + y)$ | 4. Identity |

Codes:

- | A | B | C | D |
|-------|---|---|---|
| (a) 3 | 4 | 1 | 2 |

- Ans : (a) Identity law:** $x + 0 = x$ and $x \cdot 1 = x$
Dominance law: $x + 1 = 1$ and $x \cdot 0 = 0$
Absorption law: $x(x + y) = x$
Idempotent law: $x + x = x$ and $x \cdot x = x$

(a) ISO9001 (b) RMMM
(c) CMM (d) None of these

3. The risk mitigation, monitoring and management (RMMM) plan documents all works performed as part of risk analysis and is used by the project manager as part of overall project plan.

- E-R diagram
- Data flow diagram
- Software architecture
- Software design

3. An entity relationship diagram (ERD) shows the relationships of entity sets stored in a database. An entity in this content is a component of data. In other words, ER diagrams illustrate the logical structure of database.

- Waterfall model
- Prototyping model
- Incremental model
- Evolutionary Development model

Ans : (a) 1. The **waterfall model** is a sequential (non-iterative) design process. In this model, no working software is produced until late during the life cycle.

2. The **prototyping model** is systems development method (SDM) in which a prototype (an early approximation of a final system or product) is built, tested, and then reworked as necessary until an acceptable prototype is finally achieved from which the complete system or product can now be developed.

3. The **incremental build model** is a method of software development where the product is designed, implemented and tested incrementally (a little more is added each time) until the product is finished. It involves both development and maintenance.

30. Equivalence partitioning is a _____ testing method that divides the input domain of a program into classes of data from which test cases can be derived.

- (a) White box (b) Black box
(c) Regression (d) Smoke

Ans : (b) The tendency is to relate equivalence partitioning to so called black box testing which is strictly checking a software component at its interface, without consideration of internal structure of the software.

31. Consider the following characteristics:

- (i) Correct and unambiguous
- (ii) Complete and consistent
- (iii) Ranked for importance and/or stability and verifiable
- (iv) Modifiable and Traceable

Which of the following is true for a good SRS?

- (a) (i), (ii) and (iii)
(b) (i), (iii), and (iv)
(c) (ii), (iii) and (iv)
(d) (i), (ii), (iii) and (iv)

Ans : (d) To property satisfy the basic goals, an SRS should have certain properties and should contain different types of requirements. Given, some of the desirable characteristics of on SRS (software requirement specification) and components of an SRS. A good SRS is:

- | | |
|-----------------------|---|
| 1. Correct | 2. Complete |
| 3. Unambiguous | 4. Verifiable |
| 5. Consistent | 6. Modifiable |
| 7. Traceable | 8. Ranked for importance and
and / or stability |

32. Linked Lists are not suitable for _____

- (a) Binary Search
(b) Polynomial Manipulation
(c) Insertion
(d) Radix Sort

Ans : (a) Finding middle element on array $O(1)$.
Finding mid element on linked list $O(n)$.

Therefore, recurrence relation of binary search on sorted array:

$$T(n) = T(n/2) + O(1) = \theta(\log n)$$

Recurrence relation of binary search on sorted linked list:

$$T(n) = T(n/2) + O(n) = \theta(n)$$

Since, binary search on linked taking time $\theta(n)$, so, it is inefficient on linked list.

33. What is the size of the following Union?

Assume that the size of int = 2, size of float = 4, size of char = 1

union = 1

```
{ int a ;
  float b ;
  char c ;
};
```

(a) 2

(b) 4

(c) 1

(d) 7

Ans : (b) The amount of memory required to store a structure variable is the sum of the size of all the members. On the other hand, in case of unions, the amount of memory required is always equal to that required by its largest member.

Therefore, size of union = max {2, 4, 1} = 4

34. What is the output of the following program segment?

sum (n)

```
{ if (n < 1) return n;
  else return (n + sum (n-1));
}
```

main ()

```
{ print ("%d", sum (5));
}
```

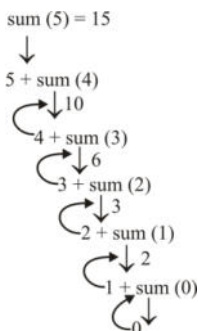
(a) 10

(b) 16

(c) 15

(d) 14

Ans : (c)



35. Assume that x and y are non-zero positive integers. What does the following program while ($x \neq 0$)

```
{
  if ( $x > y$ )
     $x = x - y$ 
```

```

else
y = y - x;
print ("%d", x);
}

```

- (a) Computes LCM two numbers
- (b) Computes GCD of two numbers
- (c) Divides large number with small number
- (d) Subtracts smaller number from large number

Ans : (b) Given algorithm is Euclidean algorithm to compute greatest common divisor (GCD) of two numbers. Function *gcd* (a, b)

While ($a \neq b$)

if $a > b$

$a = a - b$;

else

$b = b - a$;

36. Consider the following program segment:

```

d = 0;
for (i = 1; i < 31, ++i)
for (j = 1; j < 31, ++j)
for (k = 1; k < 31, ++k)
if ((i + j + k) % 3 == 0);
d = d + 1;
print ("%d", d);

```

The output will be

- (a) 9000
- (b) 3000
- (c) 90
- (d) 2700

Ans : (a) $(a + b + c) \% 3$ will be 0 if $(a + b + c)$ is a multiple of 3. This will happen in one of the following ways. All a , b and c are multiples of 3. This can only happen if a , b and c take one of the 10 values, 3, 6, 9, ..., 27, 30 independent of one another.

So, there are $10 \times 10 \times 10 = 1000$ ways this can happen. Another possibility is that a , b and c all leave a remainder 1 so that $(a + b + c)$ is evenly divisible by 3. Considering all the different possibilities and adding, we get 9000. That will be the integer that gets printed.

37. Usage of preemption and Transaction Rollback prevents_____.

- (a) Unauthorized usage of data file
- (b) Deadlock situation
- (c) Data manipulation
- (d) File preemption

Ans : (b) If there is no rule to use the system resources. Means if all the system resources are not allocated in the manner of scheduling. Then this will also create a problem for a deadlock because there is no surety that a process will release the system resources after completion, so, preemption and transaction rollback prevents deadlock situation.

38. The _____ language was originally designed as the Transformation Language for style Sheet facility.

- (a) XSTL (b) XML
(c) XQuery (d) XPath

Ans : (a) XSTL (Extensible Style sheet Language Transformations) is a language for transforming XML documents into other XML documents, or other formats such as HTML for web pages, plain text or XSL, formatting objects, which may subsequently be converted to other formats, such as PDF, Postscript and PNG. While XML (Extensible Markup Language) is a markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine readable.

39. Views are useful for _____ unwanted information, and for collecting together information from more than one relation into a single view.

- (a) Hiding (b) Deleting
(c) Highlighting (d) All of these

Ans : (a) A view is a virtual table based on the result set of an SQL statement. A view contains rows and columns, just like a real table. The field in a view are fields from one or more real tables in the database. Views can hide the complexity of data.

40. The decision tree classifier is a widely used technique for_____.

- (a) Classification (b) Association
(c) Partition (d) Clustering

Ans : (a) Decision tree classifier is a simple and widely used classification technique. It applies a straightforward idea to solve the classification problem. Decision tree classifier poses a series of carefully crafted questions about the attributes of the test record. Each time it receive an answer a follow-up question is asked until a conclusion about the class label of the record is reached.

41. Cross_tab displays permits users to view_____ of multidimensional data at a time.

- (a) One dimension (b) Two dimensions
(c) Three dimensions (d) Multidimensions

Ans : (b) A crosstab query is a type of select query which allows data to view as both horizontally and vertically so that the data can be more compact and easier to read.

42. A method to provide secure transmission of email is called_____.

- (a) TLS (b) SA
(c) IPSec (d) PGP

Ans : (d) Pretty good privacy (PGP) is an encryption program that provides cryptographic privacy and authentication for data communication. PGP is often used for signing, encryption, and decryption, e-mails, files, directories, and whole disk partitions and to increase the security of e-mail communication.

43. Thomas-write rule is_____.

- (a) Two phase locking protocol
(b) Timestamp ordering protocol

- (d) Sliding window protocol

Ans : (b) The **Thomas write rule** is a rule in timestamp ordering protocol in which write to write conflict are ignored.

44. Match the following

List-I (Process state transition)

A. Ready $\rightarrow$ Running

B. Blocked $\rightarrow$ Ready

C. Running $\rightarrow$ Blocked

D. Running $\rightarrow$ Ready

1. Request made by the process is satisfied or an event for which it was waiting occurs.

2. Process wishes to wait for some action by another process.

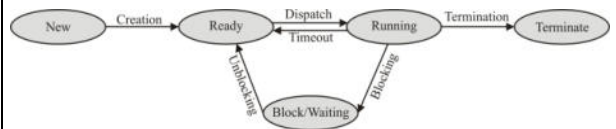
3. The process is dispatched.

4. The process is preempted.

Code:

	A	B	C	D
(a)	3	1	2	4
(b)	4	1	3	2
(c)	4	3	1	2
(d)	3	3	2	1

Ans : (a) The state diagram for a process captures its life-cycles. The states represent the execution status of the process; the transactions represent changes of execution state.



45. The hit ratio of a Translation Look Aside Bullet (TLAB) is 80%. It takes 20 nanoseconds (ns) to search TLAB and 100 ns to access main memory.

The effective memory access time is_____

- (a) 36 ns (b) 140 ns
(c) 122 ns (d) 40 ns

Ans : (b) Given, hit ratio (x) = 0, 8

TLB access time (c) = 20 nsec

Main memory access time (m) = 100 nsec

Then, effective memory access time is:

$$\text{EMAT} = x(c + m) + (1 - x)(c + 2m)$$

$$= 0.8 \times (20 + 100) + 0.2 \times (20 + 2 \times 100)$$

$$= 0.8 \times 120 + 0.2 \times 220$$

$$= 140 \text{ nsec.}$$

46. Consider the input/output (I/O) requests made at different instants of time directed at a hypothetical disk having 200 tracks as given in the following table:

Serial No.	1	2	3	4	5
Track No.	12	85	40	100	75

Time of arrival	65	80	110	100	175
------------------------	-----------	-----------	------------	------------	------------

Current head position is at track number 65. Direction of last movement is towards higher numbered tracks Current clock time is numbered tracks Current clock time is 160 milliseconds.

Head movement time per track is 1 millisecond "look" is a variant of "SCAN" disk-arm scheduling algorithm. In this algorithm, if no more I/O requests are left in current direction, the disk head reverses its direction. The seek times in Shortest Seek First (SSF) and "Look" disk-arm scheduling algorithms respectively are:

- (a) 144 and 123 milliseconds
- (b) 143 and 123 milliseconds
- (c) 149 and 124 milliseconds
- (d) 256 and 186 milliseconds

Ans : (b) At clock time 160 millisecond only request number 1 2 3 4 are there to be serviced request number 5(with track no 75) will be ready after 15 millisecond starting head position 65.

Now for SSTF 12 40 75 85 100

It will serve 85(since 75 is not come yet) first then 75 then 100 then 40 and finally 12.

So, total movements = $20 + 10 + 25 + 60 + 28 = 143$

Total time = $143 \times 1 \text{ ms} = 143 \text{ ms}$

Now for look 12 40 75 85 100 direction towards highest no. of tracks.

It will serve 85(since 75 is not come yet) first then 100 then 75 then 40 and finally 12.

So, total movement = $20 + 15 + 25 + 35 + 28$
 $= 123$ total time $123 \times 1 \text{ ms}$
 $= 123 \text{ ms}$

- 47. Assume that an implementation of Unix operating system uses i-nodes to keep track of data blocks allocated to a file. It supports 12 direct block addresses, one indirect block address and one double indirect block address. The file system has 256 bytes block size and 2 bytes for disk block address. The maximum possible size of a file this system is**
- (a) 16 MB
 - (b) 16 KB
 - (c) 70 KB
 - (d) 71 KB

Ans : (*) Given, 12 direct, one indirect and one double indirect block addresses. The file system size has 256 bytes and 2 bytes for disk block addresses. Then, maximum possible size of file in system is

$$= [12 + (2^7) + (2^7)^2] \times 256 \text{ Byte}$$

$$= 4131 \text{ KB}$$

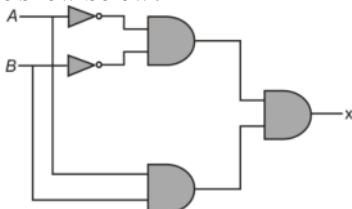
So, none option is true.

- 48. Which of the following set of Unix commands will always display "WELCOME"?**
- (a) export title = Welcome; Echo \$title
 - (b) title = WELCOME; export \$ title ; sh - c "echo \$title"

- (c) title = WELCOME; export title ; sh -c 'echo \$title'
 (d) title = WELCOME; echo \$title

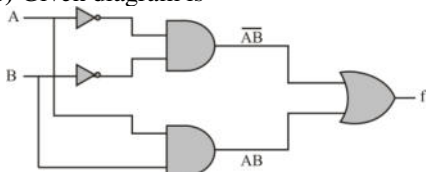
Ans : (c) title = WELCOME; export title ; sh -c 'echo \$title' will always display "WELCOME".

49. What type of logic circuit is represented by the figure show below?



- (a) XOR (b) XNOR
 (c) XAND (d) XNAND

Ans : (b) Given diagram is



Function f is:

$$f = A'B' + AB = A \odot B = A(XNOR)B$$

50. The speed up of a pipeline processing over an equivalent non-pipeline processing is defined by the ratio:

- (a) $S = \frac{nt_n}{(k+n-1)t_p}$ (b) $S = \frac{nt_n}{(k+n+1)t_p}$
 (c) $S = \frac{nt_n}{(k-n+1)t_p}$ (d) $S = \frac{(k+n-1)t_p}{nt_n}$

Where,

$n \rightarrow$ Number of tasks

$t_n \rightarrow$ Time of completion of each task

$k \rightarrow$ Number of segments of pipeline

$t_p \rightarrow$ Clock cycle time

$S \rightarrow$ speed up ratio

Ans : (a) A non-pipeline unit that performs the same operation and take a time equal to t_n to complete each task. The total time required for n tasks is nt_n .

The case where a k -segment pipeline with a clock cycle time is t_p is used to execute n tasks. The first task T_1 requires a time equal to kt_p to complete its operation since there are k segments in the pipe the remaining $(n-1)$ task emerge from the pipe at the rate of one task per clock cycle and they will be completed after a time equal to $(n-1)t_p$.

Therefore, to complete n tasks using k -segment pipeline requires $k + (n-1)$ clock cycles.

The speed-up of a pipeline processing over an equivalent non-pipeline processing is defined by the ratio.

$$S = \frac{nt_n}{(k+n-1)t_p}$$